

Table 1: Statistics of Entity Alignment datasets. ZH denotes Chinese, JA denotes Japanese, FR denotes french and EN denotes english.

Datasets	Language	Entities	Rel.	Attr.	Rel. triples	Attr. Triples	Align.
DBP _{ZH-EN}	ZH	19,388	1,701	7,780	70,414	379,684	15,000
	EN	19,572	1,323	6,933	95,142	567,755	
DBP _{JA-EN}	JA	19,814	1,299	5,681	77,214	354,619	15,000
	EN	19,780	1,153	5,850	93,484	497,230	
DBP _{FR-EN}	FR	19,661	903	4,431	105,998	528,665	15,000
	EN	19,993	1,208	6,161	115,722	576,543	
D_W_15K_V1	DBpedia	15,000	248	342	38,265	68,258	15,000
	Wikidata	15,000	169	649	42,746	138,246	
D_W_15K_V2	DBpedia	15,000	167	175	73,983	66,813	15,000
	Wikidata	15,000	121	457	83,365	175,686	
D_W_100K_V1	DBpedia	100,000	413	493	293,990	451,011	100,000
	Wikidata	100,000	261	874	251,708	687,860	
D_W_100K_V2	DBpedia	100,000	287	379	294,188	523,062	100,000
	Wikidata	100,000	32	38	400,518	749,787	

Table 2: Statistics of Knowledge Graph Alignment datasets. Mem-ST denotes memoryalpha-stexpanded, Star-SWT denotes starwars-swtor.

Datasets	Source	Inst.	Prop.	Class.	Triples.	Align.(Inst./class./prop.)
Star-SWT	Star Wars Wiki	145,033	700	269	8,246,033	1725/13/41
	The Old Republic Wiki	4,180	368	101	146,148	
Mem-ST	Memory Alpha	45,828	325	181	2,526,928	1358/15/56
	Star Trek Expanded Universe	13,426	202	283	567,386	

Statistics of Datasets

Table 1 and Table 2 summarize the statistics of the datasets used in our experiments. Following AttrGNN [2], we use the DBP15K version enriched with attribute values from the DBpedia dump (2016-10). We exclude OpenEA D-Y [5] due to its name bias, as almost all aligned entities in DBpedia and YAGO share the same names [3]. We also discard the SRPRS [1] and DWY100K [4] datasets, as they achieve perfect results using simple name-based heuristics [6].

References

- [1] GUO, L., SUN, Z., AND HU, W. Learning to exploit long-term relational dependencies in knowledge graphs. In *International conference on machine learning* (2019), PMLR, pp. 2505–2514.
- [2] LIU, Z., CAO, Y., PAN, L., LI, J., AND CHUA, T.-S. Exploring and evaluating attributes, values, and structures for entity alignment. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (2020), pp. 6355–6364.

- [3] QI, Z., ZHANG, Z., CHEN, J., CHEN, X., XIANG, Y., ZHANG, N., AND ZHENG, Y. Un-supervised knowledge graph alignment by probabilistic reasoning and semantic embedding. In *International Joint Conference on Artificial Intelligence* (2021).
- [4] SUN, Z., HU, W., ZHANG, Q., AND QU, Y. Bootstrapping entity alignment with knowledge graph embedding. In *IJCAI* (2018), vol. 18.
- [5] SUN, Z., ZHANG, Q., HU, W., WANG, C., CHEN, M., AKRAMI, F., AND LI, C. A benchmarking study of embedding-based entity alignment for knowledge graphs. *Proceedings of the VLDB Endowment* 13, 11 (2020).
- [6] ZHAO, X., ZENG, W., TANG, J., WANG, W., AND SUCHANEK, F. M. An experimental study of state-of-the-art entity alignment approaches. *IEEE Transactions on Knowledge and Data Engineering* 34, 6 (2020), 2610–2625.